

END SEMESTER EXAMINATIONS, SPRING 2025

Program: BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & ENGINEERING)		
Subject Name: DIGITAL IMAGE PROCESSING		
Subject Code: 2000015103	Semester: VI	
Duration: 2 Hours	Maximum Marks: 50	

Note: Answer the following questions as directed.

Mod. No.- Module/Topic/Unit No., **BL-** Blooms Taxonomy, **CO-** Course Outcome.

Group A

(Answer All questions)

(Each question carries 2 marks)

(5 X 2)

1. Answer the following questions:

- Write two main differences between Image Restoration and Image Enhancement?
- Can two images have the same histogram? Justify your Answer.
- Find the number of bits required to store a 256 X 256 image with a 32 gray levels?
- What is sampling and Quantization?
- How does color quantization affect the size of an image file?

Mod. No.	BL	CO
1	2	1
2	3	2
2	3	3
1	2	3
6	2	2

Group B

(Answer any Four questions out of Six)

(Each question carries 5 marks)

(4 X 5)

2. What are the fundamental steps of Digital Image Processing? Draw and explain.

Mod. No.	BL	CO
1	2	2

3. Consider the following table which represents the original pixel intensity values (r_k) and the frequency (n_k) of each unique pixel value. Perform Histogram Stretching.

Gary Level (r_k)	0	1	2	3	4	5	6	7
No. of pixels (n_k)	0	0	18	14	28	40	0	0

4. (a) What is the importance of Histogram Equalization?
 (b) Find Histogram Equalization for the Image given below:

0	3	7	7	7	2	6	2
4	3	0	2	2	7	6	2
3	3	0	7	7	0	7	1
4	3	5	5	5	5	5	5
3	1	1	2	6	0	1	1
4	1	5	7	7	7	7	7
3	0	4	7	0	7	0	7

(2 + 3)

5. Consider the following image:

5	1	3	2	7
0	3	1	1	6
0	1	4	4	2
1	6	3	2	5
3	3	4	7	5

Calculate the output image from the above image using power law transformation ($c = 1$; $\gamma = 0.2$) and logarithmic transformation ($c = 1$).

Mod. No.	BL	CO
2	4	4
2	2	2
2	3	3
4	2	3

6. Consider the following Image below:

8	7	5	6	7	7	8	8
7	7	2	3	7	6	7	8
8	6	3	4	8	6	5	8
6	7	2	2	7	7	8	7
8	7	9	5	2	2	1	3
8	6	5	6	4	3	3	2
8	8	7	7	1	3	1	3
8	7	5	6	2	2	8	4

- (a) Use region Splitting algorithm to segment the image by a threshold of 3.
- (b) Use region merging algorithm considering a threshold of 4.
7. Explain the binary thresholding technique using proper example.

Mod. No.	BL	CO
3	4	4
3	4	4

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. Explain Dilation and Erosion techniques to perform the Morphological Operations with proper example.
9. Explain the RGB color model using a proper diagram.
10. Compare and contrast global and local registration techniques in detail.

Mod. No.	BL	CO
4	3	4
6	3	3
5	2	3

Mod. No.	BL	CO
2	3	3

11. Let $I = \{0,0,1,0,0\}$ be an image. Using the mask $K = \{3,2,8\}$, perform the convolution and correlation.